Fiscal Year:	FY 2014	**Task Last Updated:**	FY 04/02/2014
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Project Title:	Development/Evaluation of a 10^-17 Yb Optical Clock for Space Applications		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Fundamental physics		
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Human Research Program Risks:	None		
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Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	80305-3337	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2011 Fundamental Physics NNH11ZTT001N
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No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JPL
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Flight Program:			
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Key Personnel Changes/Previous PI:			
COI Name (Institution):	Diddams, Scott (National Institute of Standards & Technology) Ludlow, Andrew (National Institute of Standards & Technology)		
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Task Description:	The primary goal of ESA's SOC program is to develop a high-performance optical frequency standard (clock) for operation in space with 1 x 10^-17 frequency uncertainty. NIST proposes to leverage its existing research on world-leading Yb lattice optical clocks to: (1) demonstrate a laboratory-based Yb lattice optical clock with frequency uncertainty of 1 x 10^-17 or better; (2) stabilize a space-friendly fiber frequency comb to the Yb clock signal and (3) develop a preliminary design for space-qualification by providing clock subsystem specifications. The proposed program will provide ESA with detailed information to make the most informed decisions on use of one or more optical frequency standards for the flight portion of the SOC program. Currently, ESA has identified the Sr and Yb lattice optical clocks as the most promising technologies. A Yb lattice clock has significant potential advantages over Sr in that Yb clocks can operate with solid state lasers much more likely to be practical in space operations. Several European		

Task Description:	groups are presently evaluating Sr's space prospects and Sr has thus received significant attention. For Yb, however, the most mature and high performance lattice clock resides in our lab at NIST. This Yb clock currently performs at least as well as the world's best Sr clocks, approximately 10^-16 frequency uncertainty in all cases, and continues to rapidly improve. Thus, we are uniquely qualified to play a vital role in the SOC project. For a modest investment, the proposed research program will ensure ESA has the best information and options to select the optical clock technology to meet SOC program goals, ensuring fully informed decisions affecting a much larger financial and scientific investment in the SOC program. Furthermore, by demonstrating a 10^-17 level Yb optical lattice clock, our proposed effort will achieve significant milestones currently outlined in the SOC program.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The NIST SOC program (Yb lattice clock) also directly supports the major benefits of the overall ACES program by developing a new high performance NIST optical frequency standard that will directly contribute to NIST's roles in the broader ACES program: 1. Substantial improvements in international coordination of extremely precise and accurate time and frequency information. Precision timekeeping and synchronization are crucial for such ubiquitous technologies as precision positioning and navigation, telecommunications, remote sensing, surveillance, and many other applications. The current technical limitations on international time and frequency transfer impose fundamental limits on these technologies. This project seeks to improve international time and frequency transfer by at least a factor of 10, with the potential for significantly greater improvements. Such improvements would not only directly enhance the current technologies depending on precision timing and synchronization, but would also open up new applications directly related to ultraprecise timing such as gravimetry, geodesy, magnetometry, and other measurements based on tiny perturbations in precision frequency measurements. The NIST Yb lattice clock represents a dramatic improvement in stability and accuracy compared to cesium microwave primary frequency standards. 2. This project will serve as a pathfinder demonstration of the ability to perform cold atom experiments in space. Cold atoms and molecules represent a major, growing research opportunity for a wide variety of technologies, from ultraprecision measurements of all kinds (time, gravity, magnetic fields, electrical quantities, temperature, etc.) to the potential for development of exotic new materials. Having mastered cold atom and molecule experiments in the Earth laboratory, being able to perform such experiments in space opens the benefits of precision measurements in the space environment as well as microgravity experiments not possible in the Earth-based laboratory. 3. Fundamental physics experiments. The operation of a cold atom clock in space provides the opportunity to significantly improve measurements on the limits of special and general relativity and related phenomena, such search for possible Lorentz transformation variations and search for possible time variations in fundamental constants, at levels 10 to 100 times more precise than previous experiments. Most major physics discoveries in recent decades have come from tiny variations in expected behavior, and these experiments would have profound implications on fundamental physics and cosomology.
Task Progress:	Progress in the most recent reporting period (April 2013 through March 2014) has been substantial, with major advances including: 1. Fully commissioning the second ytterbium optical lattice clock (Yb-2). 2. A series of significant improvements in both the Yb-1 and Yb-2 clocks and systems to dramatically reduce uncertainties and improve performance reliability. 3. Demonstration of unprecedented instability in comparing the Yb-1 and Yb-2 systems of 1.6 x 10^-18 with 25,000 seconds of averaging (7 hours), an order of magnitude improvement over the previous best reported stability of any frequency standard. Progress on each task is described below. 1. Fully commissioning the second ytterbium optical lattice clock (Yb-2). Both ytterbium lattice optical frequency standards (Yb-1 and Yb-2) are now reliably operating, enabling detailed evaluations of the remaining very small uncertainties that contribute to inaccuracies and instabilities in the clocks. With substantial effort to improving Yb-2, we are able to trap more than one million atoms in the new optical build up cavity for the lattice light. This is far more than is needed for normal operations (typically about 5,000 atoms), but provides enormous range to perform highly sensitive measurements on the atom-density shifts in the Yb clocks, one of the main sources of uncertainty. The Yb-2 system includes a new blackbody radiation chamber stringently designed to provide as uniform temperatures as possible for the atom environment, as Stark shifts from blackbody radiation are also one of the largest sources of uncertainty. With both Yb-1 and Yb-2 fully commissioned and operating robustly, the team was able to perform extensive measurements and intercomparisons between the systems, pushing clock instabilities well below the 1 x 10^-17 level. 2. A series of significant improvements in both the Yb-1 and Yb-2 clocks and systems to dramatically reduce uncertainties and improve performance reliability. To make measurements at the 10^-17 level and below, it is necessary to continually identify tiny uncertainties and develop methods to either reduce those uncertainties to carefully and accurately measure the frequency shifts caused by the uncertainties. Over the past year, the team has identified and addressed a large number of uncertainties in both the Yb-1 and Yb-2 systems, including both expected and unexpected sources of uncertainty. Some of the examples include: • Successful implementation of a newly designed magnetic field coil system used during the trapping phase of the measurement cycle. Abrupt shut-off of the coil current had been causing residual mechanical vibrations inducing unexpected Doppler shifts in the optical systems, leading to clock frequency shifts. An improved system of supports and slower shut-off reduced this effect by an order of magnitude. • Newly designed blackbody radiation chambers for substantially improved temperature uniformity, to reduce one of the largest sources of uncertainty, the blackbody radiation Stark shift. • Identification and implementation of new precision resistance thermometers for measuring the temperature of the blackbody radiation chambers to 25 mK or better, corresponding to blackbody radiation Stark shifts of less than 1 x

	10^-18 (assuming temperature uniformity across the chambers). • Design and implementation of room-temperature thermal shield for Yb-1 that will permit precision measurement of the blackbody radiation shifts based on temperature changes of more than 40 K around room temperature, a far greater temperature range than has been possible in other optical clocks. • Design and implementation of a new lattice build-up cavity for Yb-2. The lattice build-up cavity provides several benefits, including larger intensity variations (by about a factor of 4) for precisely measuring lattice-based Stark shifts, cleaner spatial mode for the lattice laser light (with reduced residual traveling wave component), and larger lattice volume (cross-sectional area increased by factor of 35) leading to reduced atom density and thus reduced collisional frequency shifts. • The Yb-2 lattice build-up cavity also enabled trapping more than one million cold atoms. While this amount is far more than the approximately 5,000 atoms that will typically be used in clock operations, the ability to have such a large dynamic range in density enables precision measurement of the atom density frequency shifts, reducing the density-shift uncertainty to 10^-18 or below. With these continuing improvements, confidence is high that a frequency evaluation at the 10^-17 level or below should be feasible well within the performance period.
Bibliography Type:	Description: (Last Updated: 04/05/2023)
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